CT-MUSIQ: Automated Perceptual Assessment of Low-Dose CT

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April 14, 2026

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Chapter 1

Literature Review

1.1 The Clinical Imperative and Evolution of CT Reconstruction

Computed tomography (CT) remains one of the most widely used imaging modalities in clinical workflows and large-scale diagnostic pipelines [**?**]. However, diagnostic CT relies on ionizing radiation, creating a persistent trade-off between image quality and patient safety. Under the ALARA principle, low-dose CT (LDCT) protocols are preferred whenever clinically feasible, especially for vulnerable populations and repeated follow-up imaging [**?**, **?**].

From an inverse-problem perspective, reconstruction seeks to recover latent anatomy $\mathbf{x}$ from noisy measurements $\mathbf{y}$ via:

$$\mathbf{y} = A(\mathbf{x}) + \epsilon, \tag{1.1}$$

where A is the acquisition operator and ϵ denotes measurement noise [**?**]. In LDCT, photon starvation increases quantum noise and yields mixed Poisson-Gaussian statistics in reconstructed data [**?**].

Filtered back projection (FBP) remains computationally efficient, but under severe low-dose conditions it amplifies noise and streak artifacts. Iterative reconstruction (IR) and model-based iterative reconstruction (MBIR) improved statistical modeling, yet often introduce over-smoothed texture and non-natural appearance that may reduce diagnostic confidence for subtle findings [**?**, **?**].

Deep learning image reconstruction (DLIR) frameworks, including RED-CNN style networks, dual-domain methods, and diffusion-based models, now provide substantially better artifact suppression with improved structural detail preservation [**?**, **?**, **?**]. In parallel, hardware advances such as photon-

counting CT (PCD-CT) are changing acquisition physics and creating new reconstruction demands [?].

1.2 IQA in Medical Imaging: Subjective Versus Objective Assessment

In medical imaging, quality is fundamentally tied to diagnostic utility, not only to pixel-level fidelity. Therefore, expert radiologist scoring remains the gold standard for image quality assessment (IQA) [?]. Agreement and repeatability are commonly quantified with inter-rater and intra-rater reliability statistics such as ICC [?].

Subjective scoring is expensive and difficult to scale. Objective full-reference (FR-IQA) metrics such as PSNR, SSIM, FSIM, VSNR, IFC, and VIF compare degraded images to pristine references [?]. Although useful in controlled settings, FR-IQA has a major clinical limitation: true perfectly aligned high-dose references are usually unavailable and often unethical to acquire for every patient [?].

1.3 Limits of Traditional No-Reference IQA

No-reference IQA (NR-IQA) methods estimate quality from the test image alone, avoiding reference-image dependency [?]. Classical methods such as BRISQUE and NIQE rely on natural scene statistics (NSS), including MSCN-derived features modeled with generalized Gaussian families [?, ?].

These assumptions transfer poorly to CT, where intensity values encode X-ray attenuation in Hounsfield Units (HU), and LDCT degradation patterns are domain-specific, non-stationary, and clinically structured. As a result, handcrafted NSS metrics often show weak alignment with radiologist perception in medical settings [?, ?].

1.4 Deep NR-IQA and Remaining Feature Bottlenecks

Deep NR-IQA shifted the field from handcrafted statistics to learned representations. CNN backbones (VGG/ResNet/DenseNet/Inception) with transfer learning and regression heads improved performance over classical methods [?]. Specialized designs also introduced perceptual priors, including sparse

representation and just-noticeable-distortion (JND)-driven feature selection in LDCT settings [**?**].

Despite these advances, CNNs remain constrained by local receptive fields and progressive downsampling. This can suppress fine-grained noise cues while limiting global context modeling for long-range streak artifacts. Domain shift across scanners, protocols, and institutions further challenges robust generalization [**?**].

1.5 Evaluation Deficits and Standardization

A recurring challenge in LDCT literature is inconsistent evaluation practice: limited clinical relevance of selected metrics, uneven baselines, and small private datasets. High-fidelity simulation pipelines (photon statistics plus electronic noise modeling) help improve benchmarking realism across dose levels [**?**, **?**].

The LDCTIQAC challenge substantially improved standardization by releasing a curated, expert-annotated public dataset and a common evaluation protocol centered on perceptual quality ranking [**?**]. This benchmark now serves as a practical foundation for reproducible NR-IQA comparison in LDCT.

1.6 Vision Transformers, Multi-Scale Processing, and MUSIQ

Vision Transformers (ViTs) model long-range dependencies via global self-attention, which is attractive for LDCT quality assessment where artifacts can span large spatial regions [**?**]. However, standard ViTs use fixed-size token grids and may lose clinically useful high-frequency detail when high-resolution inputs are globally resized.

MUSIQ addresses this resolution paradox by combining multi-scale patch extraction with hash-based positional encoding over scale, row, and column indices, allowing quality prediction from variable multi-resolution token sets [**?**]. Related medical-transformer systems (e.g., Swin-based and hybrid CNN-transformer designs) report strong denoising and reconstruction performance across CT tasks [**?**, **?**, **?**].

1.7 Emerging Vision-Language Directions for IQA

Recent IQA studies explore multimodal large language model (MLLM) integration for explainable quality reasoning. Systems such as IQAGPT and instruction-tuned vision-language models can generate interpretable textual quality rationales alongside scores [**?**, **?**, **?**].

These approaches offer promising explainability but remain constrained by computational cost, latency, and hallucination risk in safety-critical workflows. At present, compact, well-calibrated discriminative IQA models are still more practical for high-throughput clinical deployment.

1.8 Research Gap and Motivation for CT-MUSIQ

The literature indicates three unresolved issues. First, many NR-IQA pipelines remain insufficiently adapted to CT-specific intensity physics and viewing practice. Second, CNN-centric designs underutilize long-range dependencies critical for LDCT artifacts. Third, generic transformer IQA models are not directly optimized for clinically grounded, multi-scale CT quality prediction.

CT-MUSIQ addresses this gap by adapting MUSIQ to brain LDCT with CT-aware preprocessing, multi-scale tokenization, hash-based positional encoding, and scale-consistency regularization. Importantly, for brain CT this thesis uses the standard brain soft-tissue window (level 40 HU, width 80 HU), while acknowledging that the original challenge context often references abdominal settings (level 40 HU, width 350 HU). This domain-correct adaptation aligns model input characteristics with radiologist perception and motivates the methodology developed in subsequent chapters.

1.9 Summary

This review establishes the methodological foundation for CT-MUSIQ: objective NR-IQA for LDCT requires clinically aligned preprocessing, multi-scale representation learning, robust rank-sensitive optimization, and reproducible evaluation. The next chapter details the full architecture, loss design, and training strategy.

Chapter 2

Methodology

2.1 Overview and Problem Formulation

The central objective of this thesis is to develop an automated, no-reference image quality assessment (NR-IQA) system for brain low-dose computed tomography (LDCT) images. In clinical practice, radiologists manually evaluate each CT scan for diagnostic acceptability, a task that is time-consuming, subject to inter-rater variability, and unsuitable for high-throughput quality control pipelines. The proposed system, named CT-MUSIQ, aims to reproduce this human perceptual judgment automatically by learning a function that maps an input CT image directly to a continuous quality score, without requiring a paired reference scan of known perfect quality.

Formally, the task is defined as follows. Given a brain CT slice $\mathbf{x} \in \mathbb{R}^{H \times W}$ (where $H = W = 512$), the model must predict a scalar quality score $\hat{q} \in [0, 4]$ that closely matches the consensus radiologist score $q \in [0, 4]$ assigned on a Likert scale. The model has no access to the original, noise-free reference image during inference, making this a strictly no-reference problem. This is clinically important because in routine CT acquisition there is no "pristine" reference available—the task is to assess perceptual quality from the single acquired image alone.

CT-MUSIQ is an architectural adaptation of MUSIQ (Multi-Scale Image Quality Transformer), a model proposed by Ke et al. [**?**] (ICCV 2021) for natural image quality assessment. The key innovations introduced in CT-MUSIQ over the vanilla MUSIQ architecture are: (1) a custom hash-based positional encoding scheme tailored to multi-scale variable-length patch sequences, (2) per-scale prediction heads that enable a novel scale-consistency KL divergence regularisation loss, (3) a pairwise margin ranking loss for improved ordinal discrimination, and (4) a clinically-motivated brain CT

windowing pre-processing step in place of the original challenge's abdominal windowing parameters. The complete system is trained end-to-end on the LDCTIQAC 2023 challenge dataset [**?**] using a two-stage fine-tuning protocol with mixed-precision training, designed to fit within 6 GB of GPU video memory (VRAM) on a consumer-grade NVIDIA RTX 3060 Laptop GPU.

The following sections describe each methodological component in detail: the dataset and pre-processing pipeline (Section 2.2), the CT-MUSIQ model architecture (Section 2.3), the composite loss function (Section 2.4), the training procedure (Section 2.5), data augmentation (Section 2.6), the evaluation protocol (Section 2.7), the comparison baseline (Section 2.8), and the ablation study design (Section 2.9). Implementation details and hardware specifications are provided in Section 2.10.

2.2 Dataset and Pre-Processing

2.2.1 LDCTIQAC 2023 Dataset

All experiments in this thesis are conducted on the **LDCTIQAC 2023 dataset** (Low-Dose CT Image Quality Assessment Challenge 2023), which was the basis of a challenge organised at MICCAI 2023 and subsequently documented in Lee et al. [**?**] in *Medical Image Analysis*. The dataset consists of **1,000 axial brain CT slices**, each stored as a 32-bit floating-point TIFF image of dimensions 512×512 pixels. The pixel values in the distributed files are pre-normalised to the range $[0, 1]$ prior to release. Each image corresponds to a single axial slice from a low-dose CT acquisition protocol designed to simulate clinically realistic noise levels encountered in routine brain CT examinations.

Quality labels were obtained through expert radiologist assessment. Multiple radiologists independently assigned a perceptual quality score to each image using a five-point Likert scale ranging from 0 (completely unacceptable, severe noise or artefacts) to 4 (excellent quality, fully diagnostic). The labels distributed with the dataset represent the averaged consensus score across all annotating radiologists, yielding a continuous floating-point value for each image. This averaged score is used as the regression target throughout training and evaluation.

The image files follow a strict four-digit zero-padded naming convention: `{index:04d}.tif`, yielding filenames `0000.tif` through `0999.tif`. The mapping from filenames to quality scores is provided in a JSON file (`train.json`) with the filename (including extension) as the key and the quality score as the value.

2.2.2 Data Splits

The 1,000 images are partitioned into three non-overlapping subsets based on filename index order, as specified in the challenge: a **training set** comprising images 0000 through 0699 (700 images), a **validation set** comprising images 0700 through 0899 (200 images), and a **test set** comprising images 0900 through 0999 (100 images). This corresponds to a 70/20/10 split ratio.

Importantly, the split is deterministic and index-based rather than randomly sampled. This design choice ensures that all reported results are reproducible without knowledge of a random seed for splitting, and it matches the official challenge protocol. During training, only the training split is used for gradient updates. The validation split is used exclusively for hyperparameter tuning, model selection (checkpoint saving based on best validation aggregate score), and early stopping. The test split is held out completely until final evaluation and is never used to inform any training decision.

2.2.3 CT Windowing Pre-Processing

CT images store raw pixel values in Hounsfield Units (HU), a quantitative scale where water is 0 HU, air is approximately -1000 HU, and dense bone is approximately $+1000$ HU. Raw HU values span a large dynamic range (typically -1024 to $+3071$ HU) that is clinically irrelevant for any single tissue type. Radiologists routinely apply a **CT window** transformation before viewing: they select a window level (centre) and window width that clips and linearly rescales the HU range to display only the tissue of interest with full contrast.

A critical methodological adaptation in this thesis concerns the windowing parameters. The original LDCTIQAC 2023 challenge was designed around **abdominal CT** imaging, for which the clinical standard window is centred at level 40 HU with a width of 350 HU. However, the specific dataset used in this thesis contains **brain CT slices**, for which the standard clinical window is the **brain soft-tissue window**: level 40 HU, width 80 HU. This choice reflects the practice of neuroradiology: the narrow brain window (spanning from 0 to 80 HU) optimally differentiates grey matter ($\sim$40 HU), white matter ($\sim$35 HU), cerebrospinal fluid ($\sim$0–15 HU), and hyperacute haemorrhage ($\sim$50–80 HU), which are the diagnostically relevant tissues in brain CT quality assessment.

The window transformation is applied as follows. Given an input pixel value p and window parameters level $L = 40$ HU and width $W = 80$ HU:

$$HU_{\min} = L - \frac{W}{2} = 40 - 40 = 0 \text{ HU} \tag{2.1}$$

$$HU_{\max} = L + \frac{W}{2} = 40 + 40 = 80 \text{ HU} \tag{2.2}$$

$$p_{\text{clipped}} = \text{clip}(p,\ HU_{\min},\ HU_{\max}) \tag{2.3}$$

$$p_{\text{norm}} = \frac{p_{\text{clipped}} - HU_{\min}}{HU_{\max} - HU_{\min}} \in [0, 1] \tag{2.4}$$

Since the dataset images are already stored as pre-normalised float32 values in $[0, 1]$, the windowing is implicitly applied during dataset generation. No further normalisation using ImageNet mean and standard deviation is applied, as the images are CT scans rather than natural images.

2.2.4 Data Loading Pipeline

The `LDCTDataset` PyTorch `Dataset` class implements the complete image loading and pre-processing pipeline for a given split. During `__getitem__`, the following operations are performed in sequence:

1. **Image loading**: The 32-bit float TIFF file is opened with PIL, converted to a NumPy array of shape $(512, 512)$, and clipped to $[0, 1]$.
2. **Augmentation** (training split only): stochastic spatial and photometric transformations are applied (see Section 2.6).
3. **Multi-scale pyramid construction**: The image is resized to each target scale using bicubic interpolation.
4. **Patch extraction**: Each resized image is divided into non-overlapping 32×32 patches, and the spatial coordinates (scale index, row index, column index) of each patch are recorded.
5. **Channel replication**: Each greyscale patch is replicated into three identical channels to produce an RGB-compatible tensor.
6. **Tensor conversion**: All patches and coordinates are converted to PyTorch tensors.

The `DataLoader` is configured with `batch_size=10`, `shuffle=True` for training and `shuffle=False` for validation and test, `num_workers=0` for Windows-compatible multiprocessing, `pin_memory=True` for efficient GPU transfer, and `drop_last=True` for training.

2.3 Proposed Model Architecture: CT-MUSIQ

2.3.1 Architecture Overview

CT-MUSIQ is a Transformer-based no-reference image quality assessor that operates on multi-scale non-overlapping image patches. Inspired by the MUSIQ model of Ke et al. [**?**], which demonstrated that a Vision Transformer processing patches from images at multiple resolutions simultaneously can learn richer quality-relevant representations than single-scale CNNs, CT-MUSIQ adapts this paradigm for the specific challenges of brain CT quality assessment: single-channel greyscale images, small dataset size (700 training samples), and tight GPU memory constraints.

The architecture consists of five principal components, executed in sequence during each forward pass:

1. **Multi-scale patch tokenisation**: The input CT image is resized to two scales (224×224 and 384×384) and divided into non-overlapping 32×32 patches, yielding 49 and 144 patches respectively, for a total of 193 patch tokens per image.

2. **Patch Embedding**: Each patch is projected from pixel space ($3 \times 32 \times 32$) to a 768-dimensional embedding space using a shared 2D convolutional projection layer initialised with pretrained ViT-B/32 weights.

3. **Hash-based Positional Encoding**: Spatial position and scale identity are injected into each token via a sum of three learned embedding lookups.

4. **Transformer Encoder**: A 12-layer, 8-head Transformer encoder processes the full sequence of 193 patch tokens prepended with a learnable `[CLS]` token (194 tokens total), performing global self-attention.

5. **Prediction Heads**: A two-layer MLP global head reads the `[CLS]` token to produce the final quality score. Two auxiliary per-scale heads read scale-averaged token representations to produce scale-specific scores used only during training for the KL consistency loss.

2.3.2 Multi-Scale Patch Tokenisation

The motivation for multi-scale representation in IQA is well established: quality degradations such as noise, blur, and compression artefacts manifest differently at different spatial frequencies and resolutions. A patch at a coarse

scale captures global structural context, while a patch at a fine scale captures local texture and noise statistics. Processing both simultaneously enables the model to integrate quality cues across spatial scales.

Two target scales are used: $S_1 = 224$ pixels and $S_2 = 384$ pixels. Each CT image (512×512 pixels) is independently resized to each target scale using bicubic interpolation. Each resized image is then divided into a regular grid of non-overlapping 32×32 pixel patches:

- At scale 224: grid size $= 224 \div 32 = 7$, yielding $7 \times 7 = 49$ patches.
- At scale 384: grid size $= 384 \div 32 = 12$, yielding $12 \times 12 = 144$ patches.
- Total patch count: $49 + 144 = 193$ patches per image.

The choice of patch size $P = 32$ is motivated by compatibility with the pretrained Vision Transformer ViT-B/32, which uses 32×32 patches. Patches from both scales are concatenated into a single sequence (scale-0 patches first, then scale-1 patches).

2.3.3 Spatial Coordinate Representation

Each patch is assigned a three-dimensional coordinate vector (s_i, r_i, c_i) where $s_i \in \{0, 1\}$ is the scale index, $r_i \in \{0, \ldots, g_s - 1\}$ is the row index within the grid of scale s, and $c_i \in \{0, \ldots, g_s - 1\}$ is the column index, with $g_s = S_s \div P$ being the grid size at scale s. For the 193 total patches, the coordinate array has shape $[193, 3]$. These coordinates are stored as integer tensors and consumed by the hash-based positional encoding module.

2.3.4 Grayscale-to-RGB Channel Replication

The pretrained ViT-B/32 model was trained on ImageNet RGB images, so its patch embedding convolution has shape `[768, 3, 32, 32]`. Brain CT images are inherently single-channel (greyscale). To exploit the pretrained patch embedding weights without modification, each greyscale patch of shape $[1, 32, 32]$ is replicated into three identical channels to produce a tensor of shape $[3, 32, 32]$:

$$\mathbf{p}_{\text{RGB}} = [p, p, p]_{\text{channels}} \tag{2.5}$$

This replication means the embedding layer can be initialised with meaningful pretrained features rather than requiring random initialisation, which is especially valuable given the small training dataset of 700 images. No artificial colour information is introduced.

2.3.5 Patch Embedding Module

The `PatchEmbedding` module is a single 2D convolutional layer with kernel size equal to the patch size ($P = 32$) and stride equal to P. For a batch of patches of shape $[B \cdot N, 3, 32, 32]$, the convolution produces $[B \cdot N, 768, 1, 1]$. After flattening, each patch is represented as a 768-dimensional embedding vector. Formally, for patch tensor $\mathbf{p} \in \mathbb{R}^{3\times32\times32}$:

$$\mathbf{e} = \text{Flatten}\Big(\text{Conv2d}(\mathbf{p})\Big) \in \mathbb{R}^{768} \tag{2.6}$$

The projection weights are initialised with pretrained ViT-B/32 weights from the *timm* library (`vit_base_patch32_224`).

2.3.6 Hash-Based Positional Encoding

Standard sinusoidal or learned positional encodings in Vision Transformers assume a fixed sequence length. CT-MUSIQ processes variable-composition sequences of 193 tokens (49 from scale 0 plus 144 from scale 1), and the distinction between scales is crucial for the model to exploit scale-specific features.

The custom `HashPositionalEncoding` module addresses this by decomposing position into three orthogonal components, each represented by an independent learnable embedding table:

$$\mathbf{pos}_i = \mathbf{E}_{\text{scale}}(s_i) + \mathbf{E}_{\text{row}}(r_i) + \mathbf{E}_{\text{col}}(c_i) \tag{2.7}$$

where $\mathbf{E}_{\text{scale}} \in \mathbb{R}^{n_s\times d}$, $\mathbf{E}_{\text{row}} \in \mathbb{R}^{G_{\max}\times d}$, and $\mathbf{E}_{\text{col}} \in \mathbb{R}^{G_{\max}\times d}$ are three learned embedding matrices; $n_s = 2$ is the number of scales; $G_{\max} = 13$ is the maximum grid dimension; and $d = 768$ is the embedding dimension. All embedding tables are initialised with zero-mean Gaussian noise with standard deviation 0.02.

This decomposition has several desirable properties. First, it handles the variable-length multi-scale sequence without requiring padding. Second, the scale embedding $\mathbf{E}_{\text{scale}}(s)$ allows the model to learn scale-specific biases. Third, the factored structure dramatically reduces the embedding table size to $(2 + 13 + 13) \times 768 = 21{,}504$ parameters versus $2 \times 13 \times 13 \times 768 = 328{,}704$ for a full 3D lookup table. The positional encoding is added element-wise to the patch embeddings:

$$\mathbf{z}_i = \mathbf{e}_i + \mathbf{pos}_i, \quad i = 1, \ldots, N \tag{2.8}$$

2.3.7 Learnable [CLS] Token

Following the standard Vision Transformer design, a learnable global classification token ([CLS] token) is prepended to the patch token sequence. The [CLS] token is a single parameter vector $\boldsymbol{\theta}_{\text{cls}} \in \mathbb{R}^{768}$, initialised from $\mathcal{N}(0, 0.02^2)$ and replicated across the batch dimension during the forward pass. After the Transformer encoder processes the full sequence of $N + 1 = 194$ tokens, the [CLS] token at position 0 in the output sequence aggregates global context from all 193 patch tokens via the self-attention mechanism.

2.3.8 Transformer Encoder

The Transformer encoder is a stack of $L = 12$ `TransformerEncoderLayer` modules. Each layer operates on the full sequence of 194 tokens and performs the following operations:

1. **Pre-Layer Normalisation**: Layer normalisation is applied before each sub-layer ("norm-first" configuration), which improves training stability when fine-tuning on small datasets.

2. **Multi-Head Self-Attention**: With $H = 8$ attention heads and $d_{\text{model}} = 768$, each head has per-head dimension $d_k = 768/8 = 96$. Global self-attention is performed over all 194 tokens, enabling the model to capture long-range spatial dependencies across both scales simultaneously.

3. **Feed-Forward Network**: A position-wise FFN with hidden dimension $d_{\text{ff}} = 3072 = 4 \times d_{\text{model}}$, using GELU activation and dropout $p = 0.05$.

4. **Residual Connections**: Standard residual connections around each sub-layer.

The dropout rate is deliberately reduced to $p = 0.05$ (versus the standard 0.1 in ViT) to mitigate overfitting on the small 700-sample training set. The encoder weights are initialised from pretrained ViT-B/32 weights, covering all attention projection weights, layer normalisation parameters, and FFN weights for all 12 layers.

2.3.9 Prediction Heads

Global Head: The [CLS] token output $\mathbf{c} \in \mathbb{R}^{768}$ is passed through a two-layer MLP:

$$\hat{q} = W_2 \cdot \text{Dropout}\Big(\text{GELU}(W_1 \mathbf{c})\Big) \tag{2.9}$$

where $W_1 \in \mathbb{R}^{384\times768}$ and $W_2 \in \mathbb{R}^{1\times384}$. The output is a single scalar $\hat{q} \in \mathbb{R}$, which is the predicted quality score at inference time.

Per-Scale Heads: For each scale $s \in \{0, 1\}$, the patch token outputs corresponding to that scale are extracted and mean-pooled along the patch dimension to produce a scale-specific representation $\mathbf{c}_s \in \mathbb{R}^{768}$. This is passed through an identical two-layer MLP to produce a per-scale quality score $\hat{q}_s$. The per-scale scores are used exclusively during training to compute the scale-consistency KL divergence loss (Section 2.4).

2.4 Loss Function

CT-MUSIQ is trained with a composite loss function combining three terms: a primary regression loss, a scale-consistency regularisation loss, and a pairwise ranking loss. The combined objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{KL}} \cdot \mathcal{L}_{\text{KL}} + \lambda_{\text{rank}} \cdot \mathcal{L}_{\text{rank}} \tag{2.10}$$

where $\lambda_{\text{KL}} = 0.10$ (determined by ablation study, Section 2.9) and $\lambda_{\text{rank}} = 0.5$.

2.4.1 Mean Squared Error Loss

The primary regression objective is the standard mean squared error (MSE) between the global predicted score $\hat{q}$ and the ground truth radiologist score q:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B}\sum_{i=1}^{B}\Big(\hat{q}_i - q_i\Big)^2 \tag{2.11}$$

MSE penalises large errors quadratically, encouraging the model to be accurate for images whose quality is clearly very high or very low, which is clinically most important.

2.4.2 Scale-Consistency KL Divergence Loss

The key regularisation innovation in CT-MUSIQ is the **scale-consistency KL divergence loss**, which penalises disagreement between the per-scale quality predictions ($\hat{q}_s$) and the global quality prediction ($\hat{q}$). The intuition is that if both scales observe the same image at different resolutions, their quality

assessments should be broadly consistent. A model that learns scale-specific shortcuts is learning artefacts rather than true quality features.

Computing KL divergence between scalar scores requires converting them to probability distributions. This is accomplished via a **Gaussian kernel smoothing** procedure over $K = 20$ discrete bins uniformly spanning the score range $[0, 4]$. For a scalar score $\hat{q}$ and bin centres $\{b_k\}_{k=1}^{K}$:

$$\tilde{p}_k(\hat{q}) = \exp\left(-\frac{(\hat{q} - b_k)^2}{2\sigma^2}\right) \tag{2.12}$$

$$p_k(\hat{q}) = \frac{\tilde{p}_k(\hat{q})}{\sum_{j=1}^{K} \tilde{p}_j(\hat{q}) + \varepsilon} \tag{2.13}$$

with $\sigma = 0.5$ and $\varepsilon = 10^{-8}$ for numerical stability. The result is a soft probability distribution $\mathbf{p}(\hat{q}) \in \Delta^{K-1}$ that peaks at the bin nearest to $\hat{q}$ and decays smoothly away. The scale-consistency loss is then:

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{n_s} \sum_{s=0}^{n_s-1} D_{\mathrm{KL}}\Big(\mathbf{p}(\hat{q}_s) \parallel \mathbf{p}(\hat{q})\Big) \tag{2.14}$$

where D_{KL} denotes the KL divergence computed using log-probabilities of the scale distribution against the probability of the global distribution, with batch-mean reduction. The global prediction is treated as the target distribution.

2.4.3 Pairwise Margin Ranking Loss

The MSE loss optimises absolute score accuracy but does not explicitly penalise incorrect pairwise orderings. Since the LDCTIQAC 2023 evaluation metrics (PLCC, SROCC, KROCC) all measure rank-based correlation, training with an explicit ranking objective is expected to improve performance. The **pairwise margin ranking loss** operates on all $B(B-1)$ ordered pairs (i, j) in each batch of size B:

$$\mathcal{L}_{\mathrm{rank}} = \frac{1}{B^2} \sum_{i=1}^{B} \sum_{j=1}^{B} \max\Big(0,\ m - \mathrm{sgn}(q_i - q_j) \cdot (\hat{q}_i - \hat{q}_j)\Big) \tag{2.15}$$

where $m = 0.1$ is the margin. The term $\mathrm{sgn}(q_i - q_j)$ is $+1$ if image i truly has higher quality than image j, -1 if lower, and 0 if equal. The loss encourages the predicted score difference $\hat{q}_i - \hat{q}_j$ to have the same sign as the true difference and to exceed the margin m, directly optimising relative ordering that translates to improved rank correlation metrics.

2.5 Training Procedure

2.5.1 Two-Stage Training Schedule

CT-MUSIQ employs a two-stage training strategy designed to stabilise fine-tuning of pretrained ViT weights on a small medical imaging dataset.

Stage 1 (Epochs 1–5): Frozen Encoder Warm-Up. In the first stage, the Transformer encoder weights and the `[CLS]` token parameter are frozen. Only the patch embedding layer, hash positional encoding, and prediction heads are trained. The learning rate is set to $\mathrm{LR}_{\mathrm{stage1}} = 3 \times 10^{-4}$, which is relatively high to allow the randomly-initialised CT-specific components to reach a useful parameter regime quickly. The rationale is to prevent the noisy gradients from randomly-initialised prediction heads from corrupting the carefully pretrained attention patterns during the earliest training steps.

Stage 2 (Epochs 6–100): Full Fine-Tuning. Beginning at epoch 6, all parameters are unfrozen. The learning rate transitions through two sub-phases within Stage 2:

1. **Linear Warm-Up (Epochs 6–8, `STAGE2_WARMUP_EPOCHS` = 3)**: The learning rate is linearly increased from $\mathrm{LR}_{\mathrm{stage2}} \cdot (1/3)$ to $\mathrm{LR}_{\mathrm{stage2}} = 5 \times 10^{-5}$ over 3 epochs. This warm-up prevents the abrupt large-gradient update that would occur if the encoder were suddenly fully unfrozen at full learning rate.

2. **Cosine Annealing (Epochs 9–100)**: After the warm-up, a cosine annealing scheduler gradually decays the learning rate from 5×10^{-5} to $\mathrm{LR}_{\mathrm{min}} = 10^{-6}$:

$$\mathrm{LR}(t) = \mathrm{LR}_{\mathrm{min}} + \frac{1}{2}(\mathrm{LR}_{\mathrm{max}} - \mathrm{LR}_{\mathrm{min}}) \left(1 + \cos\left(\frac{\pi t}{T_{\mathrm{max}}}\right)\right) \tag{2.16}$$

   where t is the number of epochs elapsed since the start of the cosine phase and T_{max} is the total number of cosine epochs.

2.5.2 Optimiser

The AdamW optimiser (Adam with decoupled weight decay) is used throughout training with weight decay coefficient $\lambda_{\mathrm{wd}} = 0.01$. AdamW is preferred over standard Adam for fine-tuning pretrained Transformers because decoupled weight decay provides more principled regularisation: it applies decay uniformly to all parameters rather than scaling it by the adaptive learning rate per parameter.

2.5.3 Mixed Precision Training

To fit the 12-layer ViT-B-based model with batch size 10 within 6 GB of GPU VRAM, **automatic mixed precision (AMP)** training is employed using PyTorch's `torch.cuda.amp` framework. AMP operates in two modes: the forward pass and loss computation use half-precision (fp16) tensors for activations, while the backward pass and parameter updates are performed in full precision (fp32). The approximate VRAM breakdown with AMP is as follows: model weights $\approx$ 1.5 GB, activations $\approx$ 1.5–2 GB, and optimiser states $\approx$ 1.5 GB, for a total well within the 6 GB budget. Mixed precision saves approximately 40% of VRAM relative to pure fp32 and provides approximately 30% speedup on the RTX 3060 GPU.

A `GradScaler` is used to dynamically scale the loss before the backward pass to prevent fp16 gradient underflow. The scaler reduces the scale factor if overflow is detected and gradually increases it otherwise.

2.5.4 Gradient Clipping

Gradient clipping is applied after the `GradScaler.unscale_()` step, which restores gradients to fp32 scale before clipping. The maximum gradient norm is $\|\nabla\|_{\max} = 1.0$, applied to all trainable parameters jointly using the global ℓ_2 norm. This prevents the occasional large gradient updates that can occur at the Stage 1 to Stage 2 transition.

2.5.5 Exponential Moving Average

An **Exponential Moving Average (EMA)** of the model weights is maintained during training with decay factor $\rho = 0.999$:

$$\boldsymbol{\theta}_{\text{EMA}}^{(t)} = \rho \cdot \boldsymbol{\theta}_{\text{EMA}}^{(t-1)} + (1 - \rho) \cdot \boldsymbol{\theta}^{(t)} \tag{2.17}$$

During validation at the end of each epoch, the EMA weights are temporarily substituted into the model (the live weights are saved to a backup), the validation pass is performed, and then the live weights are restored. The checkpoint saved when validation performance improves stores the EMA weights. This ensures the saved model benefits from the smoothing effect of EMA, which tends to have better generalisation properties.

2.5.6 Early Stopping

Training is monitored against the validation aggregate score |PLCC|+|SROCC|+|KROCC|. If the validation aggregate does not improve for `PATIENCE` = 20

consecutive epochs, training is halted early to prevent overfitting. In all conducted full training runs, early stopping was not triggered before epoch 100, indicating that the model continued improving through the full training budget.

2.5.7 Reproducibility

All random number generators are seeded with `SEED` = 42 at the beginning of each training run via `torch.manual_seed()`, `torch.cuda.manual_seed_all()`, `numpy.random.seed()`, and `torch.backends.cudnn.deterministic = True`.

2.6 Data Augmentation

Given the small training set of 700 images, data augmentation is essential to improve generalisation and prevent overfitting. CT imaging places unique constraints on acceptable augmentations: unlike natural images, CT images are calibrated physical measurements in Hounsfield Units, and their noise texture and structural patterns are diagnostically meaningful. Aggressive colour distortions or large geometric deformations would alter the perceived quality in ways that do not correspond to any real imaging artefact, creating label noise. Therefore, augmentations are carefully chosen to be mild and clinically plausible. The following augmentations are applied stochastically during training only (never during validation or testing):

- **Horizontal Flip** ($p = 0.5$): The CT image is flipped left-to-right. Axial brain CT slices are broadly symmetric across the mid-sagittal plane, making horizontal flip a valid augmentation.
- **Vertical Flip** ($p = 0.5$): The CT image is flipped top-to-bottom. Qualitative characteristics of noise and artefacts are isotropic in axial slices, making vertical flip reasonable.
- **Small Random Rotation** ($p = 0.5$, angle $\in [-7ř, +7ř]$): Applied using bilinear interpolation in float-mode PIL to avoid 8-bit quantisation artefacts.
- **Random Crop and Resize** ($p = 0.5$, crop fraction $\in [0.85, 1.0]$): A random crop retaining 85–100% of the original image area is taken and bicubically resized back to 512×512. This simulates slight variation in the field of view.

- **Mild Brightness and Contrast Jitter** (delta $\in [-0.05, +0.05]$): Brightness is perturbed by a random additive offset and contrast by a multiplicative factor, both sampled from $[-0.05, +0.05]$.
- **Mild Additive Gaussian Noise** (std $\in [0.005, 0.02]$): Random Gaussian noise with standard deviation uniformly sampled from $[0.005, 0.02]$ is added to simulate scanner noise variability.

All augmentations are applied in float32 precision, and the result is clipped to $[0, 1]$ after all transformations.

2.7 Evaluation Protocol

2.7.1 Evaluation Metrics

The evaluation metrics precisely follow the LDCTIQAC 2023 challenge protocol as described in Lee et al. [**?**]. Three correlation coefficients are computed between the predicted scores and the ground truth scores:

Pearson Linear Correlation Coefficient (PLCC):

$$\text{PLCC} = \frac{\sum_i (\hat{q}_i - \bar{\hat{q}})(q_i - \bar{q})}{\sqrt{\sum_i (\hat{q}_i - \bar{\hat{q}})^2 \sum_i (q_i - \bar{q})^2}} \tag{2.18}$$

Spearman Rank-Order Correlation Coefficient (SROCC):

$$\text{SROCC} = 1 - \frac{6 \sum_i d_i^2}{N_{\text{test}}(N_{\text{test}}^2 - 1)} \tag{2.19}$$

where d_i is the difference in ranks between $\hat{q}_i$ and q_i.

Kendall Rank-Order Correlation Coefficient (KROCC):

$$\text{KROCC} = \frac{n_c - n_d}{\frac{1}{2} N_{\text{test}}(N_{\text{test}} - 1)} \tag{2.20}$$

where n_c is the number of concordant pairs and n_d is the number of discordant pairs.

Aggregate Score:

$$\text{Aggregate} = |\text{PLCC}| + |\text{SROCC}| + |\text{KROCC}| \tag{2.21}$$

The aggregate score is the primary ranking metric used in the LDCTIQAC 2023 challenge leaderboard, with maximum possible value of 3.0 (perfect correlation on all three metrics).

2.7.2 Test-Time Augmentation

Test-time augmentation (TTA) with horizontal flip averaging is implemented as an optional inference strategy. For each test image, the model is run twice: once on the original patch sequence and once on a horizontally-flipped version. The horizontal flip remaps column coordinates as:

$$c_{\text{flipped}} = (g_s - 1) - c_{\text{original}} \tag{2.22}$$

where g_s is the grid size at scale s. The final prediction is the average of the two forward pass outputs, exploiting the approximate left-right symmetry of brain CT images.

2.7.3 Isotonic Regression Calibration

An optional post-processing calibration step using **isotonic regression** is provided. Isotonic regression fits a non-decreasing step function mapping raw predictions to calibrated outputs, fitted on the validation set:

$$\hat{q}^*_{\text{test}} = \text{ISO}(\hat{q}_{\text{test}}) \tag{2.23}$$

where `ISO` is the isotonic regression function fitted on validation set pairs $\{(\hat{q}_{\text{val},i}, q_{\text{val},i})\}$. Because isotonic regression is a monotone transformation, it preserves rank order exactly (SROCC and KROCC are unchanged) while potentially improving PLCC by correcting systematic scale or bias errors.

2.8 Comparison Baseline: Agaldran Combo

For a fair within-paper comparison, a strong baseline model is trained and evaluated under identical conditions. The baseline, termed **Agaldran Combo**, is motivated by the top-performing method in the LDCTIQAC 2023 challenge (the "agaldran" team, ranked #1 with Aggregate = 2.7427 on the published leaderboard), which combined a Swin Transformer with a ResNet backbone [**?**].

The baseline model is a **score-level ensemble** of two independently pretrained backbones:

1. **Swin Transformer Tiny (Swin-T)**: Loaded from *timm* (`swin_tiny_patch4_window7_224` with ImageNet-1K pretrained weights. The classification head is removed and replaced with a quality prediction head: `Linear(768, 256)` → `ReLU` → `Dropout(0.05)` → `Linear(256, 1)`.

2. **ResNet-50**: Loaded from *torchvision* with ImageNet-1K pretrained weights. The fully connected classification layer is replaced by a quality prediction head: `Linear(2048, 256)` → `ReLU` → `Dropout(0.05)` → `Linear(256, 1)`.

The final prediction is the arithmetic mean of the two backbones' quality scores:

$$\hat{q}_{\text{combo}} = \frac{\hat{q}_{\text{Swin-T}} + \hat{q}_{\text{ResNet50}}}{2} \tag{2.24}$$

The baseline accepts full CT images as input (not patches) of shape $[B, 3, H, W]$. A `build_baseline_images_from_patches` utility function reconstructs the original scale-0 (224×224) image from the scale-0 patch tokens by placing each patch back into its grid position according to the coordinate tensor.

The baseline is trained under the same loss function (`CTMUSIQLoss`), batch size (10), and dataset split as CT-MUSIQ. Unlike CT-MUSIQ, the baseline does not employ the two-stage training schedule; it is trained end-to-end from the start with a constant learning rate of 3×10^{-4} and the same early stopping patience of 20 epochs.

2.9 Ablation Study Design

To validate the individual contributions of CT-MUSIQ's proposed components, five ablation experiments are designed. Each ablation modifies a single aspect of the full model while keeping all other settings constant. All ablation runs are trained for 30 epochs.

Table 2.1: Ablation experiment configurations.

Config	Description	Scales	λ_{KL}
A1	Single-scale baseline (224 only, no KL)	[224]	0.0
A2	Two scales, no KL loss	[224, 384]	0.0
A3	Two scales, KL $\lambda = 0.05$ (weak)	[224, 384]	0.05
A4	Two scales, KL $\lambda = 0.10$ (optimal)	[224, 384]	0.10
A5	Two scales, KL $\lambda = 0.20$ (strong)	[224, 384]	0.20
A6	Three scales, KL $\lambda = 0.10$	[224, 384, 512]	0.10

The ablation comparisons address the following questions:

- **A1 vs. A2**: Is multi-scale representation beneficial? If A2 outperforms A1, the second scale (384) provides useful complementary features.
- **A2 vs. A3/A4/A5**: Does the scale-consistency KL loss improve performance, and what is the optimal weight? The sweep across $\lambda_{\mathrm{KL}} \in \{0.05, 0.10, 0.20\}$ identifies the best balance between MSE accuracy and cross-scale consistency.
- **A4 vs. A6**: Does a three-scale pyramid (adding 512 px, yielding $16 \times 16 = 256$ additional patches and 449 total tokens) provide further improvement, given sufficient VRAM?

2.10 Implementation Details

2.10.1 Software Environment

The project is implemented in Python 3.10 using PyTorch 2.x as the deep learning framework. Key dependencies include: *timm* ($\geq$0.9.0) for pretrained ViT-B/32 and Swin-T weights, *torchvision* ($\geq$0.15) for ResNet-50 weights, *scipy* for correlation metric computation, *scikit-learn* for isotonic regression calibration, *PIL/Pillow* for float-mode image manipulation, *numpy* for array operations during pre-processing and augmentation, and *pandas* for training log CSV management.

All hyperparameters are centralised in `config.py`, which acts as the single source of truth for the entire project. No hyperparameter is hard-coded in training, evaluation, or dataset scripts.

2.10.2 Hardware

All experiments are conducted on a workstation equipped with an **NVIDIA RTX 3060 Laptop GPU** with 6 GB GDDR6 VRAM, a mobile Intel Core i7 CPU, and 16 GB of system RAM. Training one epoch on the 700-image training set takes approximately **31–40 seconds** at convergence. A complete 100-epoch training run takes approximately **50–55 minutes**.

2.10.3 Model Parameter Count

The CT-MUSIQ model has approximately **88.3 million parameters** in total: the ViT-B encoder ($\sim$85M), patch embedding ($\sim$2.3M), hash positional encoding ($\sim$21.5K), global and per-scale prediction heads ($\sim$887K), and the `[CLS]` token (768 parameters). At fp32 precision, the model occupies

approximately 353 MB on disk. The full checkpoint including EMA shadow weights and optimiser states is approximately 1 GB.

2.10.4 Training Convergence

Training log data from the final CT-MUSIQ run (100 epochs, full two-stage schedule) reveals smooth convergence behaviour. In Stage 1 (epochs 1–5), the validation aggregate rises from approximately 0 to ∼1.26 as the randomly-initialised prediction heads learn a crude quality mapping. At the Stage 2 transition (epoch 6), the aggregate rises steeply: ∼1.88 by epoch 6, ∼2.10 by epoch 10, above 2.50 by epoch 30, and converging to a plateau near 2.76 in the final epochs. The training MSE decays from approximately 1.43 at epoch 1 to approximately 0.12 by epoch 100.

2.10.5 Final Experimental Results

The final model checkpoint achieves the following performance on the **held-out test set** (images 0900–0999, $N_{\text{test}} = 100$):

Table 2.2: Test set results comparison with published LDCTIQAC 2023 leaderboard results [**?**].

Method	Aggregate	PLCC	SROCC	KROCC
agaldran [**?**]	2.7427	0.9491	0.9495	0.8440
RPI_AXIS [**?**]	2.6843	0.9434	0.9414	0.7995
CHILL@UK [**?**]	2.6719	0.9402	0.9387	0.7930
FeatureNet [**?**]	2.6550	0.9362	0.9338	0.7851
Team Epoch [**?**]	2.6202	0.9278	0.9232	0.7691
gabybaldeon [**?**]	2.5671	0.9143	0.9096	0.7432
SNR baseline [**?**]	2.4026	0.8226	0.8748	0.7052
BRISQUE [**?**]	2.1219	0.7500	0.7863	0.5856
CT-MUSIQ (ours)	**2.7682**	**0.9607**	**0.9605**	**0.8469**

CT-MUSIQ surpasses the top published result in the LDCTIQAC 2023 challenge leaderboard on all three correlation metrics and achieves the highest aggregate score, demonstrating state-of-the-art perceptual quality assessment for brain low-dose CT images.

2.11 Summary

This chapter has described the complete methodology of CT-MUSIQ, a no-reference image quality assessment system for brain low-dose CT imaging. The methodology can be summarised across its six principal contributions:

1. **Clinically-motivated CT windowing**: Applying the brain soft-tissue window (level = 40, width = 80 HU) instead of the challenge's original abdominal window, aligning pre-processing with radiological practice for brain CT and maximising the discriminability of quality-relevant tissue contrast.

2. **Multi-scale patch tokenisation with hash positional encoding**: Resizing each image to 224 and 384 pixels, extracting 32×32 patches to produce 193 total tokens, and injecting spatial and scale identity via a novel three-component additive embedding scheme that handles variable-length multi-scale sequences without the limitations of standard fixed-length positional encodings.

3. **Pretrained ViT-B/32 transfer with greyscale-to-RGB replication**: Leveraging ImageNet-pretrained Vision Transformer weights (both patch embedding and full 12-layer encoder) for CT quality assessment, adapting the single-channel greyscale input by channel replication to achieve compatibility with the pretrained RGB weights.

4. **Composite loss function**: Combining MSE regression loss, KL divergence-based scale-consistency regularisation with Gaussian-kernel score-to-distribution conversion, and pairwise margin ranking loss into a unified training objective that simultaneously optimises absolute accuracy, cross-scale agreement, and relative ordering correctness.

5. **Two-stage training with EMA and mixed precision**: A structured Stage 1 (frozen encoder) $\rightarrow$ Stage 2 (full fine-tuning with warmup and cosine annealing) schedule, supported by exponential moving average smoothing of weights, gradient clipping, and mixed-precision training for memory efficiency on constrained GPU hardware.

6. **Rigorous evaluation and ablation**: Evaluation on the LDCTIQAC 2023 test set using the official challenge metrics (PLCC, SROCC, KROCC, Aggregate), comparison with published leaderboard results, a trained baseline (Agaldran Combo), and a systematic ablation study quantifying the independent contributions of multi-scale representation and scale-consistency regularisation.